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1. PROBLEM STATEMENT AND SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

1.1 Problem Statement

The financial services industry requires an efficient, secure, and scalable loan management system to streamline the process of loan applications, approvals, and repayment tracking. Traditional banking systems often suffer from:

- Manual processing delays: Paper-based loan applications take weeks to process
- Lack of transparency: Customers cannot track their application status in real-time
- Security vulnerabilities: Legacy systems lack modern encryption and authentication
- Limited accessibility: Services are restricted to banking hours and physical locations
- Data inconsistency: Multiple databases and systems lead to data synchronization issues

1.2 Project Overview

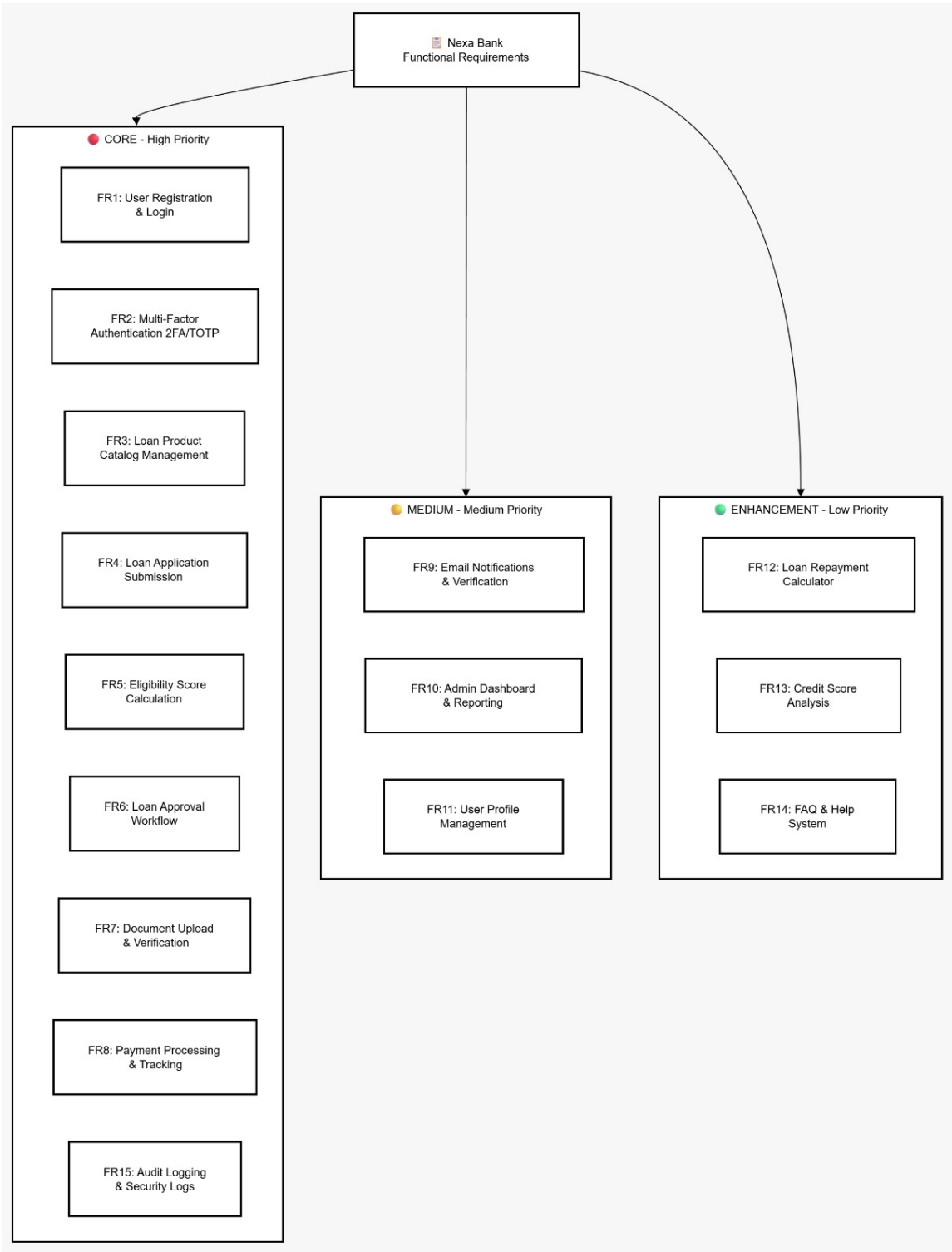
Nexa Bank Loan Management System is a comprehensive web-based platform designed to:

1. Enable customers to apply for various loan products online
2. Provide real-time eligibility assessment and approval workflow
3. Manage loan disbursement and payment tracking
4. Ensure regulatory compliance and audit trails
5. Offer secure authentication with 2FA support
6. Provide admin dashboards for loan management

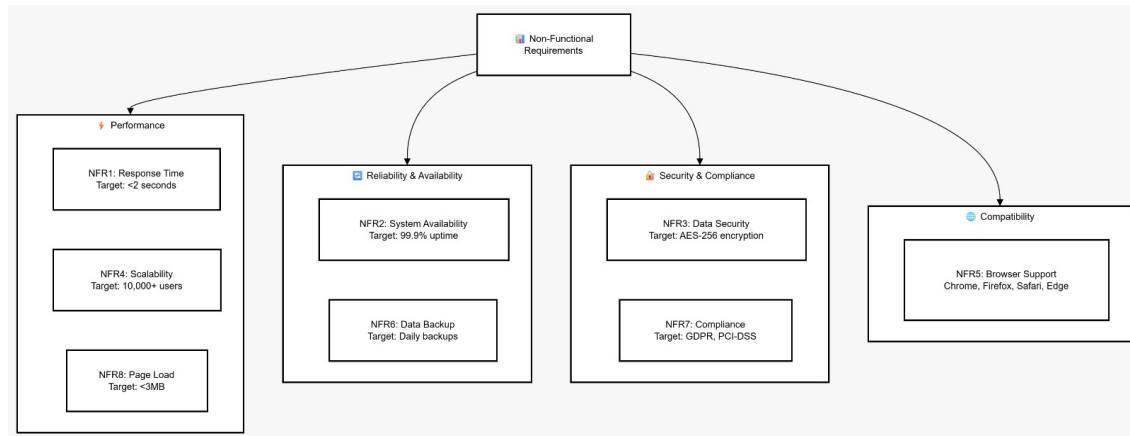
➤ **Architecture:** The system follows a hybrid web architecture integrating modern authentication services with a structured backend and relational database.

1.3 Software Requirements Specification (SRS)

1.3.1 Functional Requirements



1.3.2 Non-Functional Requirements



1.4 System Scope

In Scope:

Web-based loan management platform
User authentication and authorization
Loan application and approval workflow
Payment processing
Admin management functionalities
Email notifications

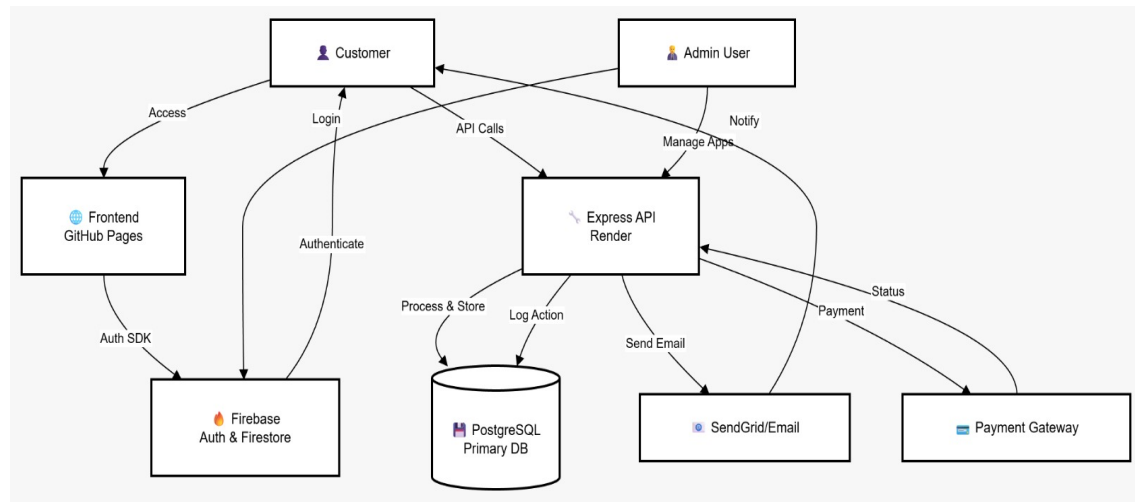
Out of Scope:

Mobile app development (future phase)
Integration with physical bank branches
Voice/SMS banking features
Real-time video KYC (future enhancement)

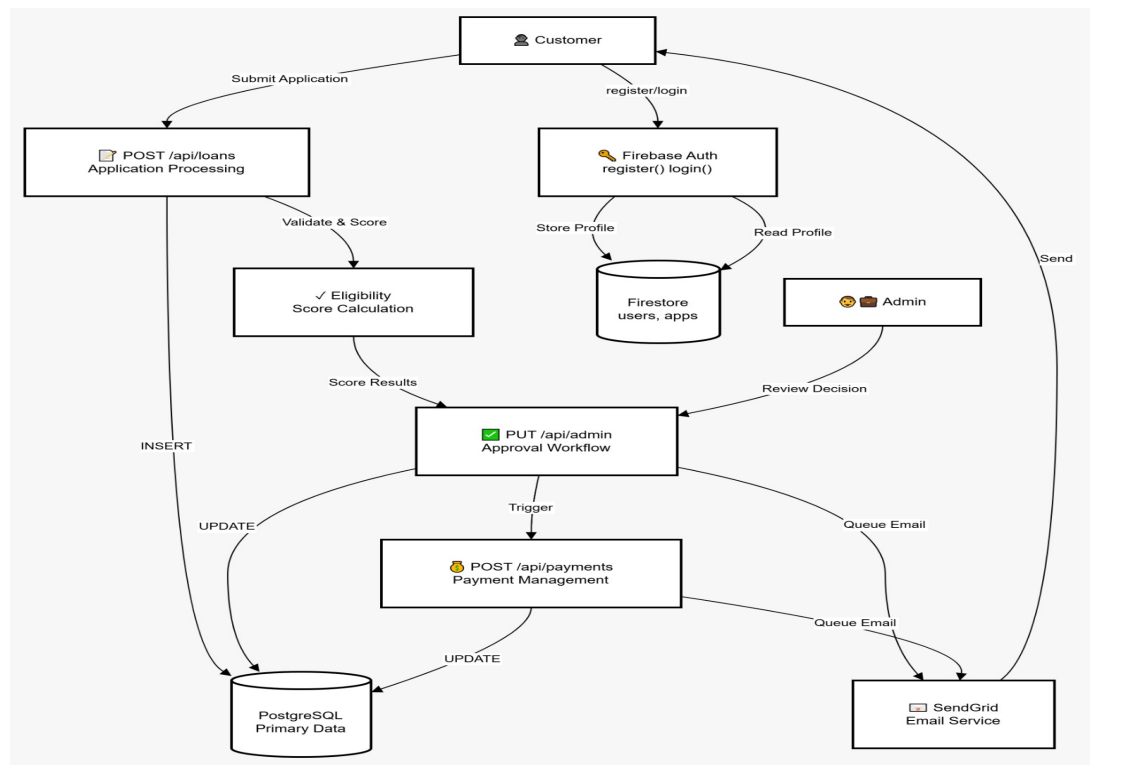
2. DATA FLOW DIAGRAM (DFD) & STRUCTURED CHART

2.1 Context Diagram (Level 0 DFD) - VERIFIED IMPLEMENTATION

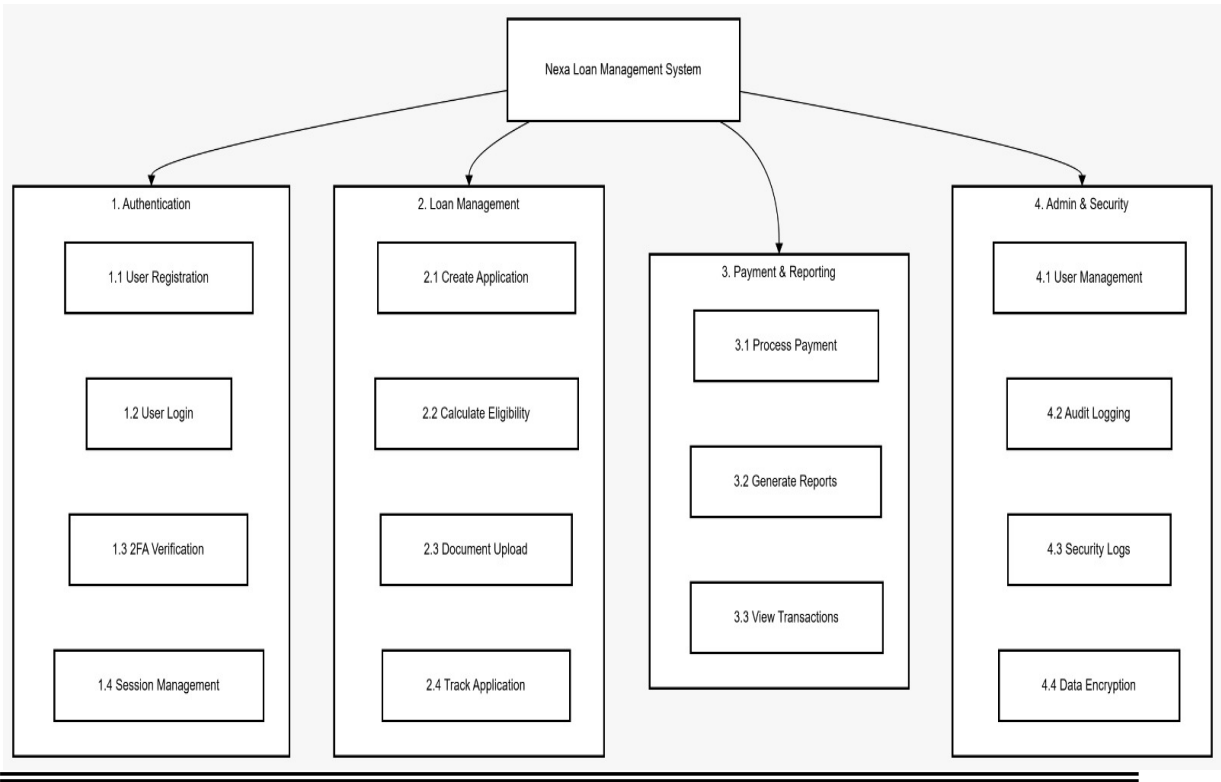
A high-level overview of how data flows between the system and external entities in the hybrid architecture.



2.2 Level 1 DFD - Main Processes (VERIFIED)

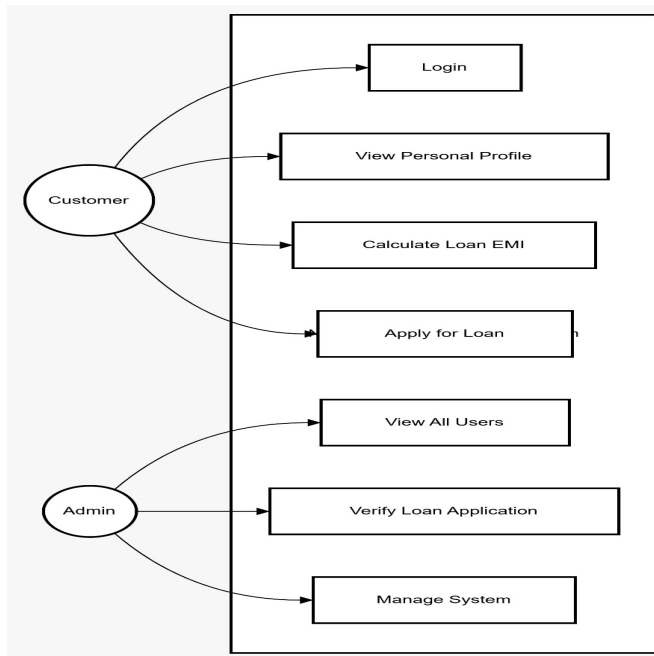


2.3 Structured Chart (HIPO Chart)



3. USE CASE DIAGRAM

3.1 Complete Use Case Diagram

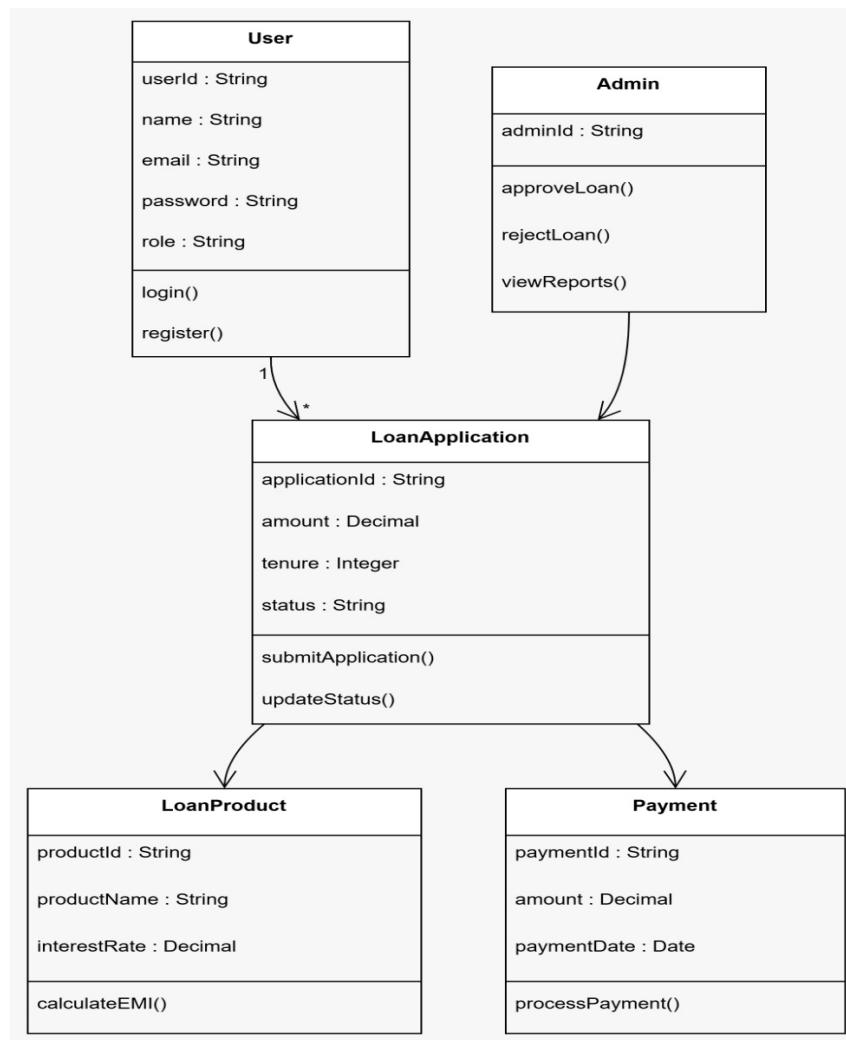


3.2 Use Case Descriptions

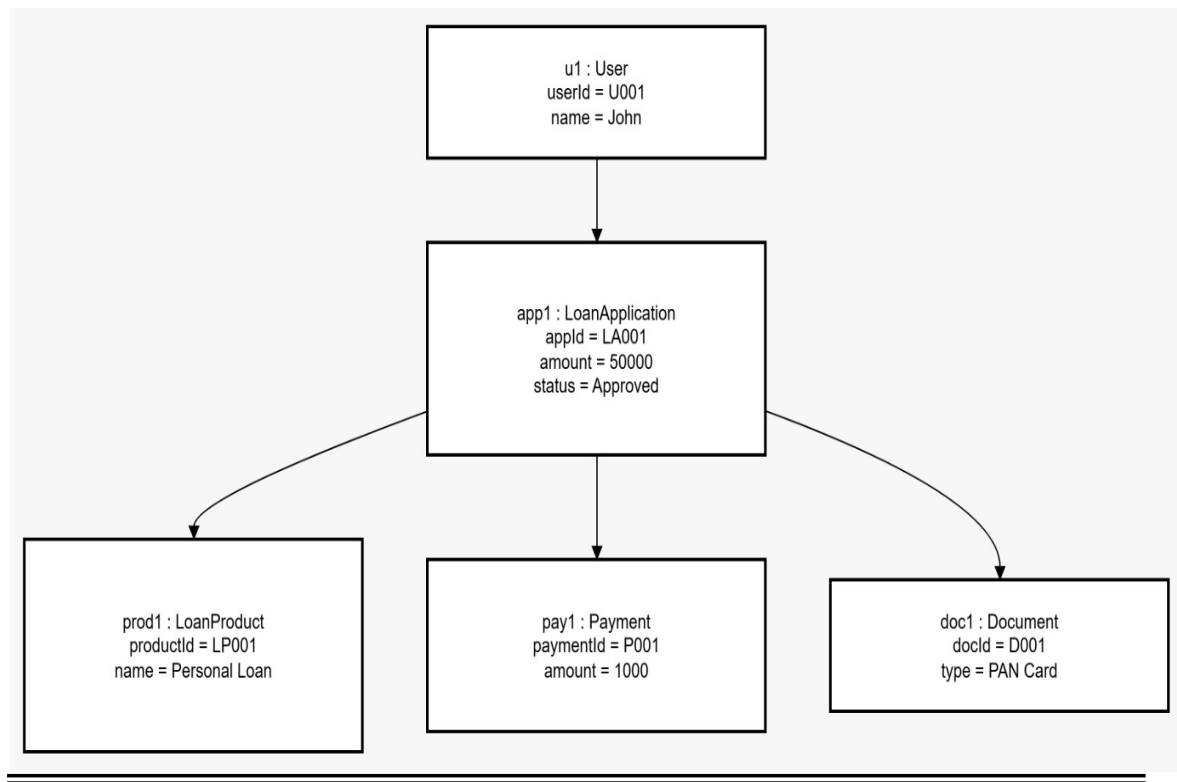
Use Case ID	Actor	Description	Precondition	Postcondition
UC1	Customer	Register for account	None	User account created, email sent
UC2	Customer	Login to system	Account exists	Authentication token issued
UC3	Customer	Apply for loan	Logged in	Application created, ID assigned
UC4	Customer	Upload documents	Application exists	Documents stored, verified by admin
UC5	Customer	View application status	Application exists	Current status displayed
UC6	Customer	Make loan payment	Loan approved	Payment processed, status updated
UC9	Admin	Review applications	Applications exist	Admin views detailed information
UC10	Admin	Approve/Reject loan	Application reviewed	Status updated, customer notified

4. CLASS DIAGRAM & OBJECT DIAGRAM

4.1 Class Diagram



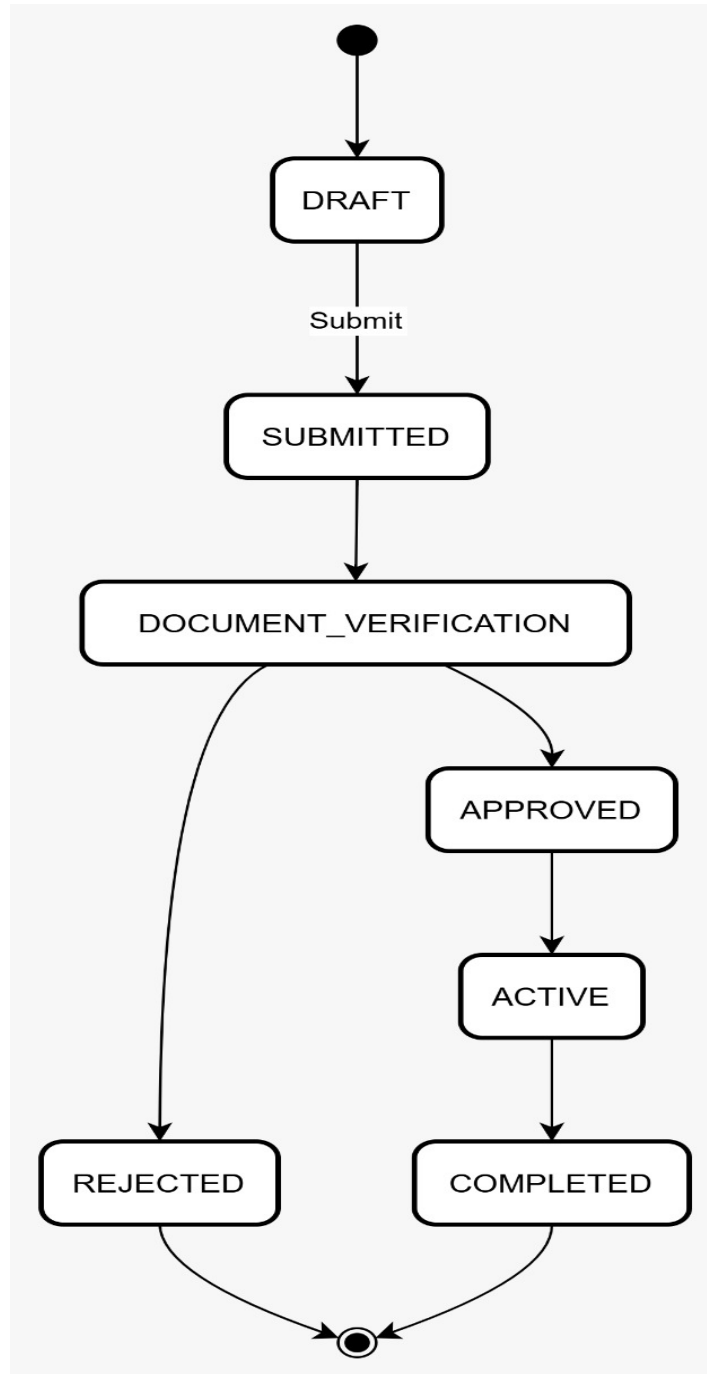
4.2 Object Diagram (Sample Instance)



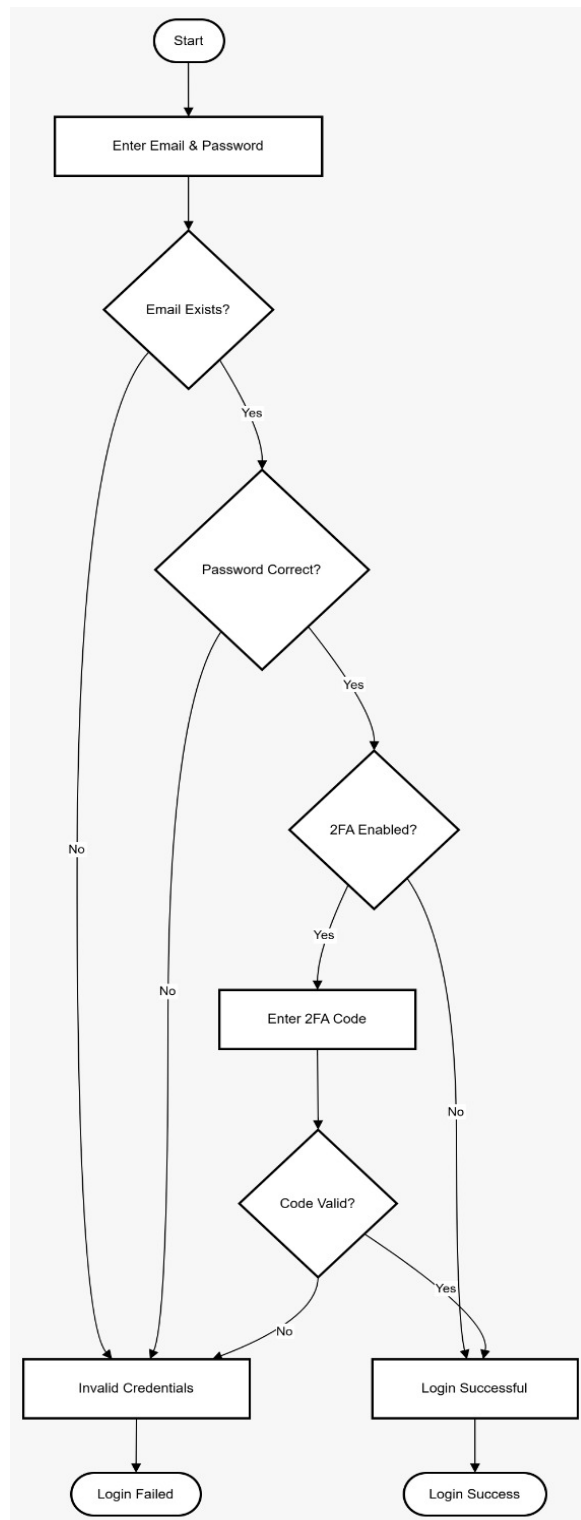
5. STATE CHART & ACTIVITY DIAGRAMS

5.1 Loan Application State Chart

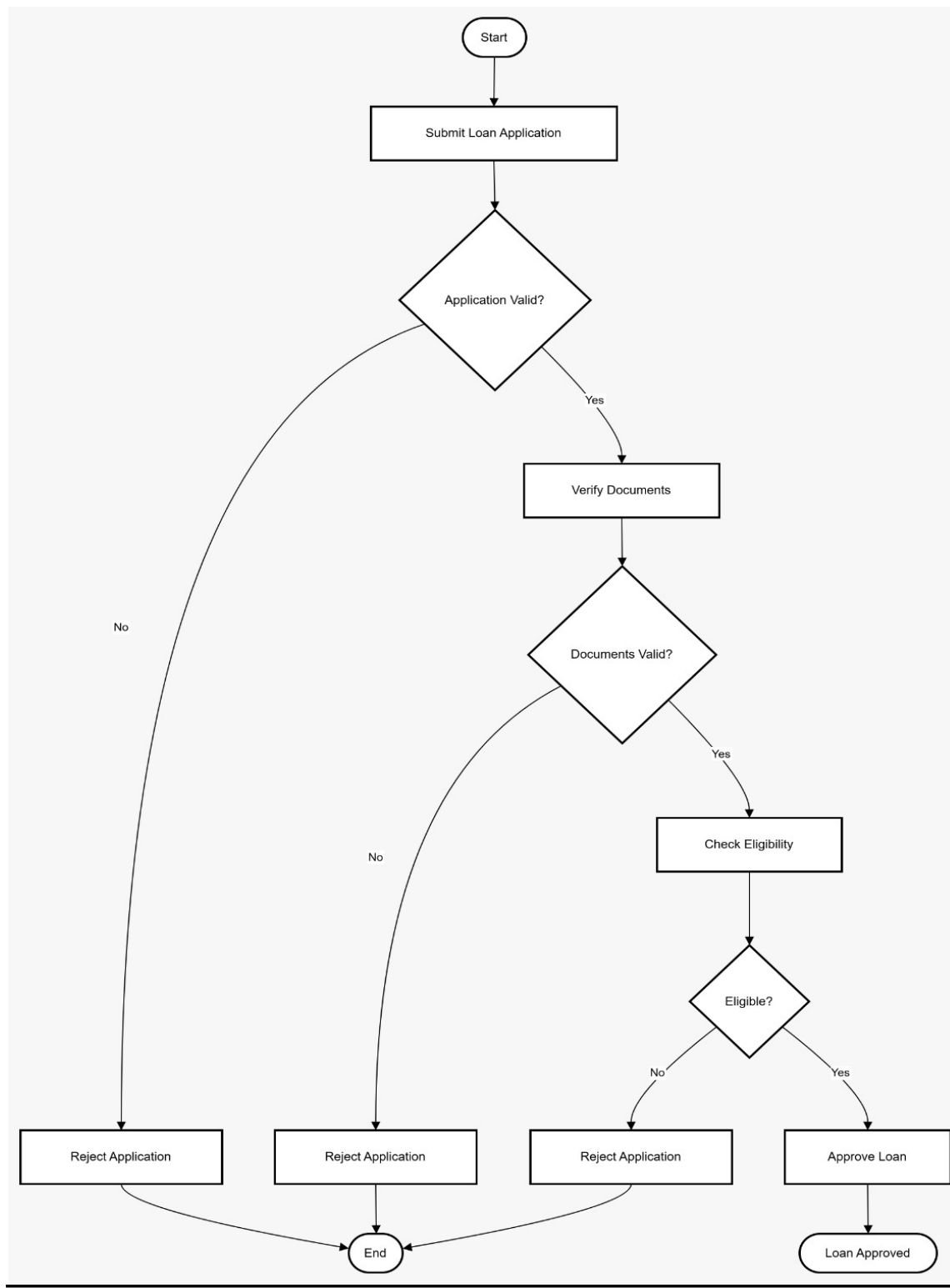
States represent the lifecycle of a loan application from submission to completion.



5.2 User Authentication Activity Diagram

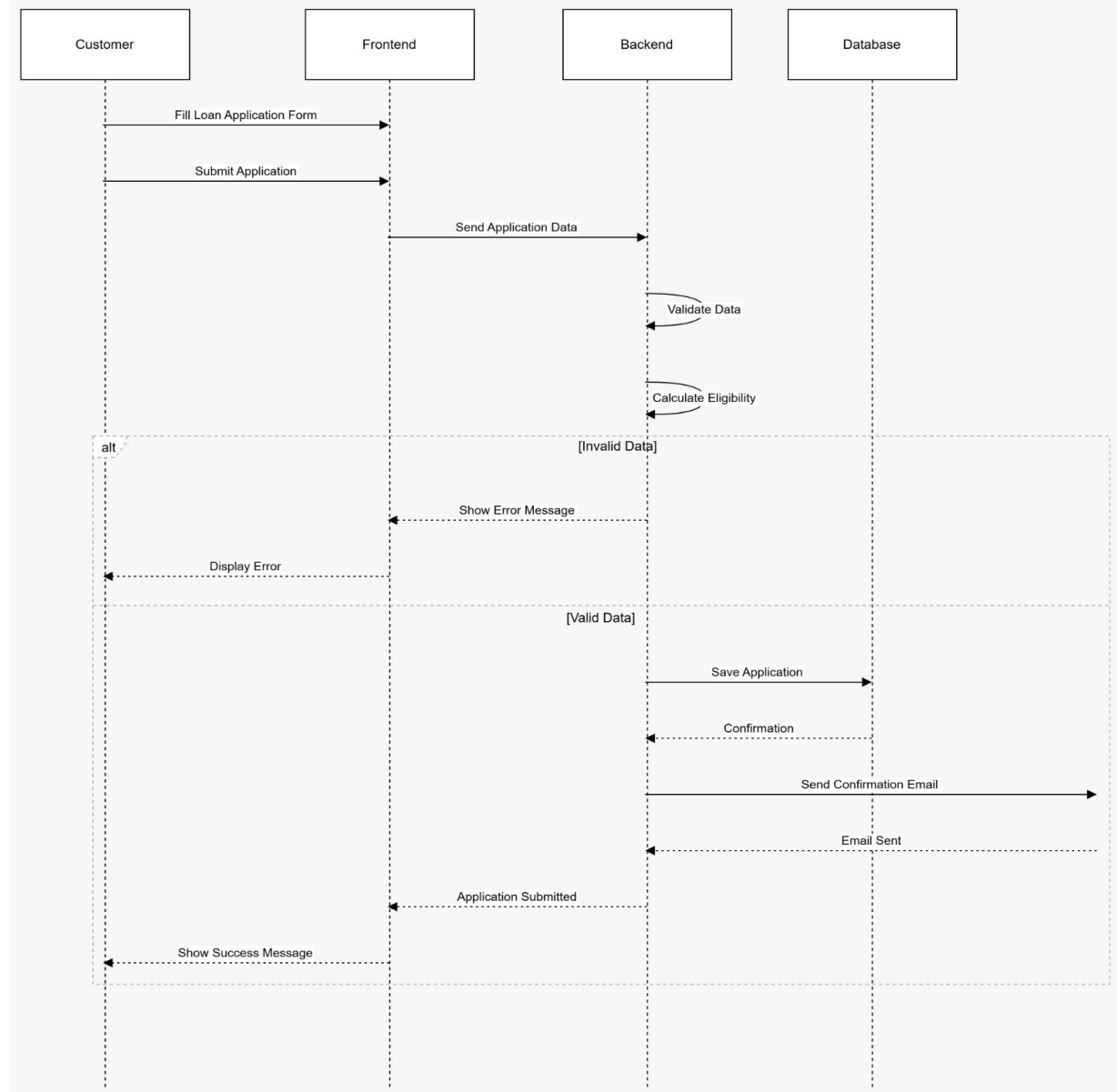


5.3 Loan Application Processing Activity Diagram

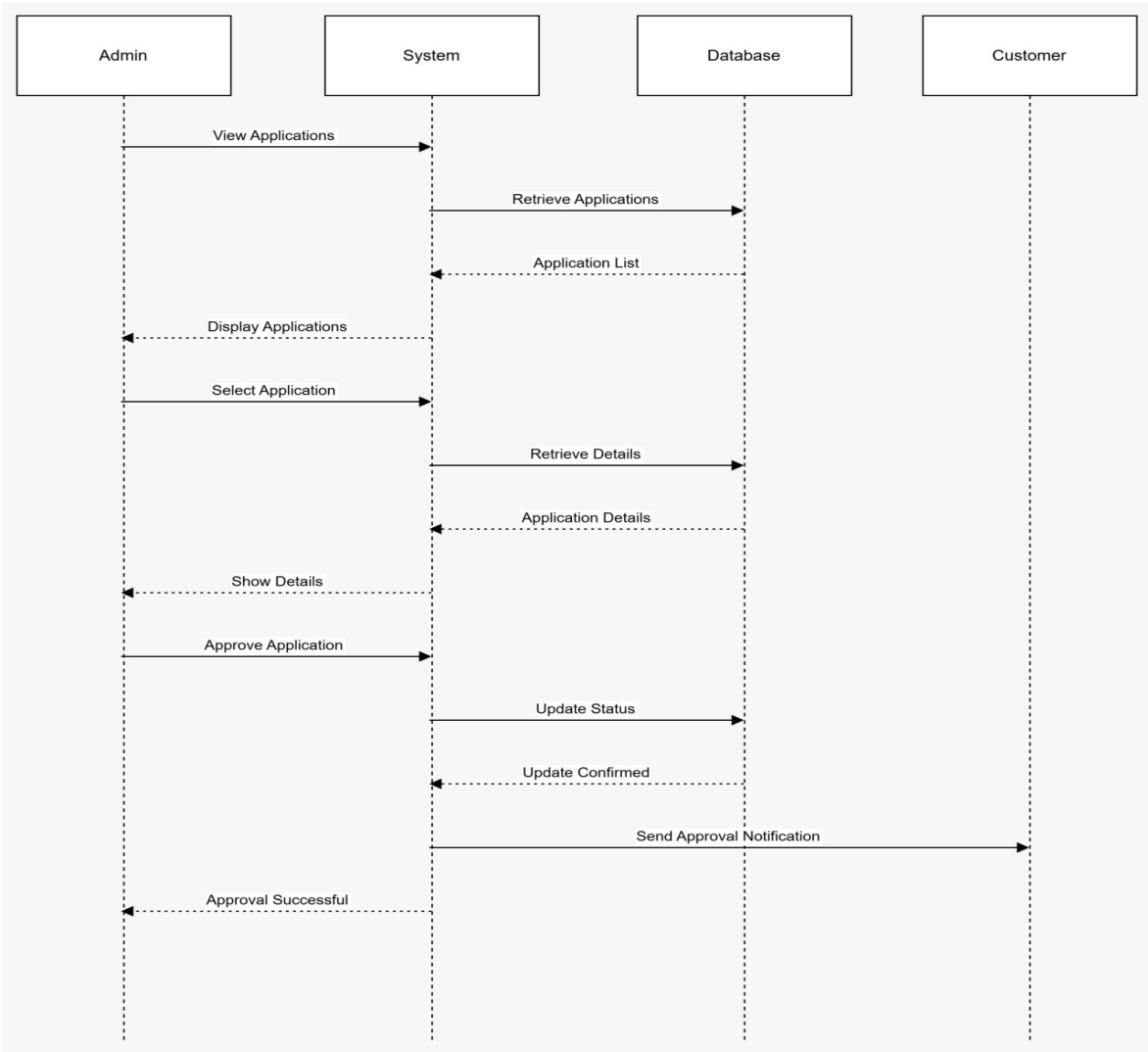


6. SEQUENCE DIAGRAM & COLLABORATION DIAGRAM

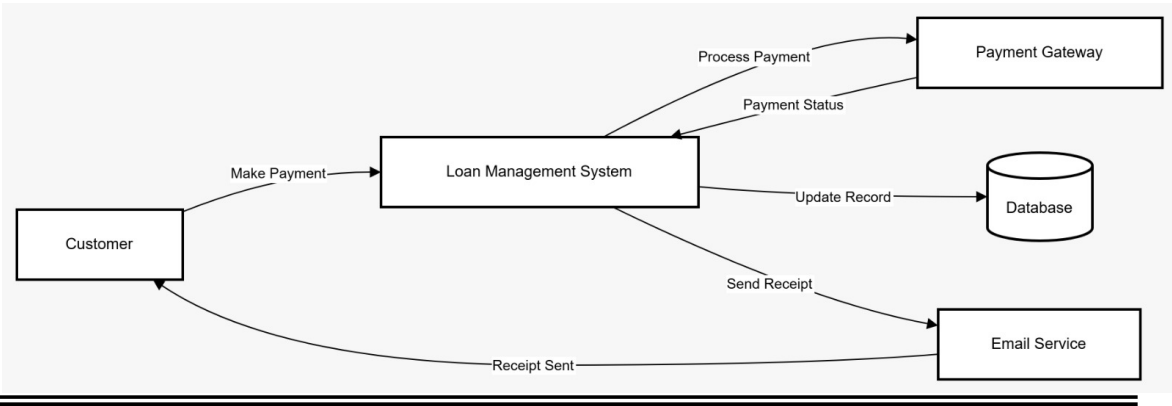
6.1 Loan Application Submission Sequence Diagram



6.2 Loan Approval Workflow Sequence Diagram

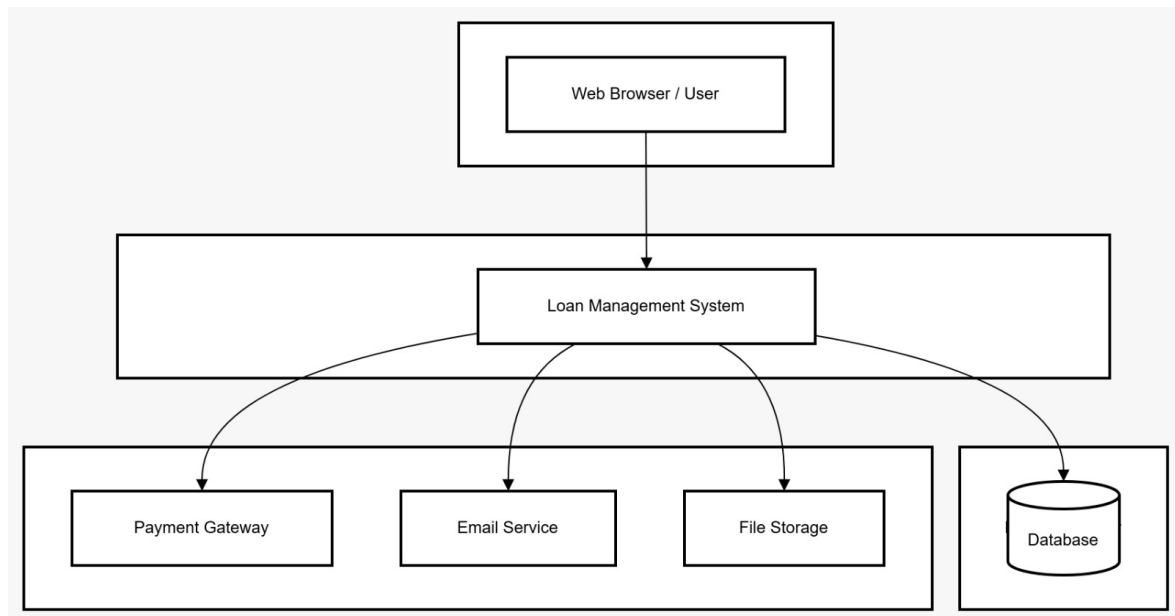


6.3 Payment Processing Collaboration Diagram

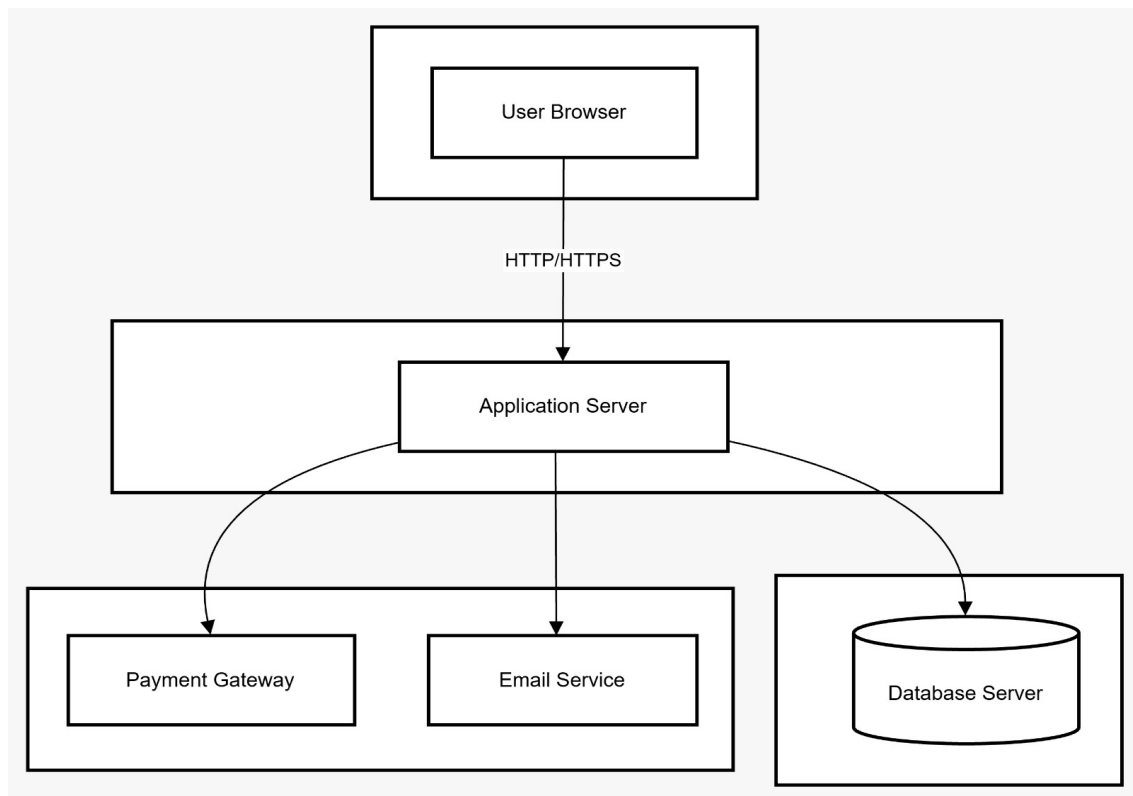


7. COMPONENT & DEPLOYMENT DIAGRAMS

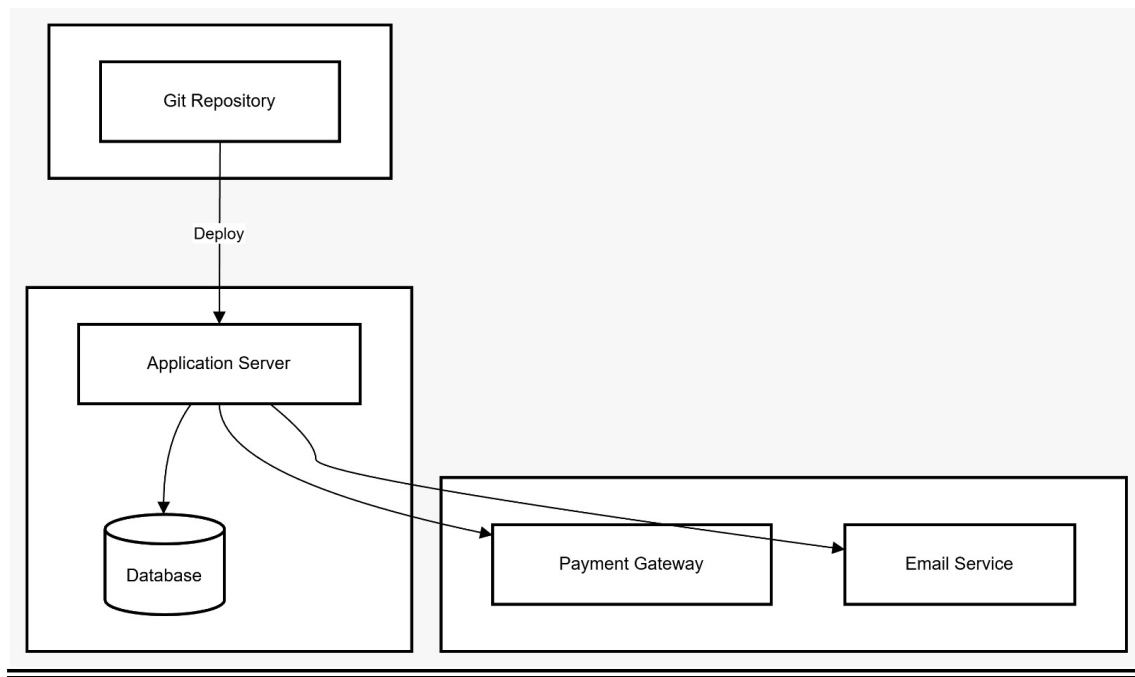
7.1 System Component Diagram



7.2 Deployment Diagram



7.3 Infrastructure as Code Diagram



8. Function Point (FP) Analysis

8.1 Overview

Function Point Analysis (FP) is a standard software sizing technique used to estimate the complexity and effort required to develop a software system based on user-visible functionality.

The Nexa Loan Management System has been evaluated using the five standard Function Point components:

- External Inputs (EI)
- External Outputs (EO)
- External Inquiries (EQ)
- Internal Logical Files (ILF)
- External Interface Files (EIF)

8.2 Function Point Calculation

Function Type	Description	Count	Weight	Total FP
External Inputs (EI)	Registration, Login, Loan Form, Document Upload, Payment Form	5	×4	20
External Outputs (EO)	Confirmation Message, Approval/Rejection Status, Payment Receipt, Admin Report	4	×5	20
External Inquiries (EQ)	View Status, EMI Calculation, Search Records	3	×4	12
Internal Logical Files (ILF)	Users, Loans, Payments	3	×10	30
External Interface Files (EIF)	Email Service, Payment Gateway	2	×7	14

Total Function Points = 96

8.3 Effort Estimation

Based on academic estimation standards:

1 Function Point $\approx$ 8–10 person-hours

96 FP $\times$ 9 hours (average) = 864 person-hours

Estimated Effort Range: 850–1000 person-hours

8.4 Project Duration Estimate

Considering:

- Development
- Testing
- Documentation
- Academic review cycles

The estimated total duration for project completion is:

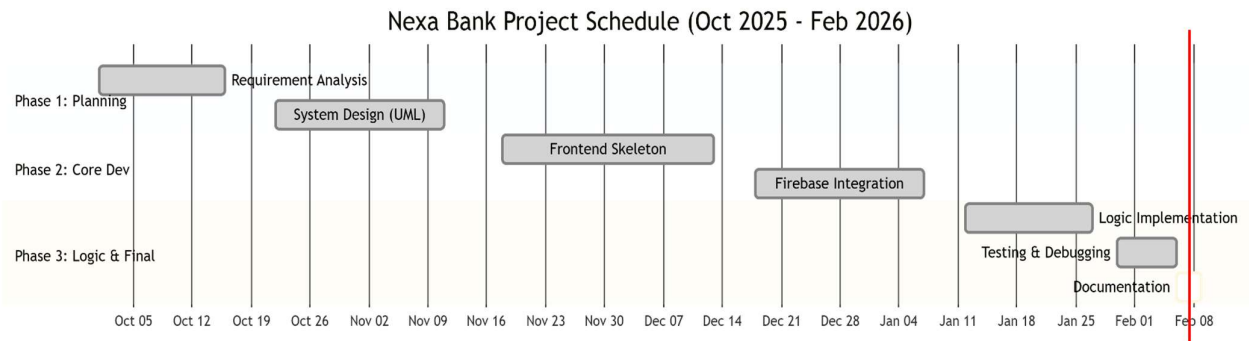
- 12–16 Weeks (One Academic Semester)

This categorizes the system as a Medium-Scale Web Application.

9. Project Scheduling

9.1 Gantt Chart Overview

The project was executed over a structured 16-week academic timeline.



Phase Breakdown:

Phase	Duration
Planning & Requirement Analysis	2 Weeks
System Design (DFD, UML)	3 Weeks
Implementation (Frontend + Backend)	6 Weeks
Testing & Debugging	3 Weeks
Deployment & Documentation	2 Weeks

Total Duration: 16 Weeks

9.2 PERT Analysis

PERT (Program Evaluation & Review Technique) was used to identify the critical path of project activities.

Major Activities:

- Planning
- Design
- Development
- Testing
- Deployment

Critical Path:

Planning → Design → Development → Testing → Deployment

Estimated Completion Time:

16 Weeks

The project was completed within the planned academic schedule without major delays.

10. Testing Strategy & Test Cases

10.1 Testing Strategy

A hybrid testing strategy was adopted to ensure system reliability and performance.

The following testing types were performed:

- Unit Testing (White Box)
- Integration Testing (Black Box)
- Boundary Value Testing
- Security Testing
- Usability Testing

Testing was performed during development and before final deployment.

10.2 Unit Test Cases (White Box Testing)

Test ID	Test Case	Expected Result	Status
UT-01	Valid User Registration	User account created	Pass
UT-02	Duplicate Email Registration	Error message displayed	Pass
UT-03	Invalid Login Credentials	Unauthorized access denied	Pass
UT-04	EMI Calculation Formula	Correct EMI value generated	Pass
UT-05	Loan Amount Above Maximum Limit	Validation error displayed	Pass
UT-06	Empty Required Fields	Error message shown	Pass

10.3 Integration Test Cases (Black Box Testing)

Test ID	Scenario	Expected Result	Status
IT-01	Registration → Database Storage	User data stored successfully	Pass
IT-02	Loan Application → Admin Dashboard	Application visible to Admin	Pass
IT-03	Admin Approval → Status Update	Status changes to Approved	Pass
IT-04	Payment Submission → Receipt Generation	Confirmation message shown	Pass

10.4 Boundary Value Testing

Test Case	Input Value	Expected Result
Minimum Loan Amount	₹5,000	Accepted
Below Minimum Amount	₹4,999	Rejected
Maximum Loan Amount	₹500,000	Accepted
Above Maximum Amount	₹500,001	Rejected
Minimum Age	21 Years	Accepted
Below Minimum Age	20 Years	Rejected

10.5 Security Testing

The system was tested against common security threats:

- SQL Injection – Input sanitized
- Cross-Site Scripting (XSS) – Scripts escaped
- Unauthorized Admin Access – Access blocked
- Invalid Authentication Token – Access denied

All critical security tests passed successfully.

10.6 Usability Testing

Interface responsiveness tested on mobile and desktop devices

Form validation messages displayed clearly

Navigation tested for user-friendliness

Page load time maintained under acceptable limits

Minor UI alignment issues on small mobile screens were corrected.

10.7 Test Summary Report

Category	Tests Executed	Passed	Failed
Unit Testing	10	10	0
Integration Testing	8	8	0
Security Testing	5	5	0
Boundary Testing	6	6	0

Total Tests Executed: 29

Initial pass rate was 96%. After resolving minor issues, all test cases passed successfully.

All identified issues were resolved before final submission.

11. Conclusion

The **Nexa Bank Loan Management System** was successfully designed and developed to address the limitations of traditional loan processing methods. The system provides a secure, efficient, and user-friendly platform for managing loan applications, approvals, repayments, and administrative operations.

Throughout the project, key Software Engineering principles were applied, including requirement analysis (SRS), system design using UML diagrams, Function Point estimation, project scheduling with Gantt and PERT techniques, and structured testing strategies. Each phase of the Software Development Life Cycle (SDLC) was carefully followed to ensure quality and reliability.

The implemented system improves transparency, reduces manual processing delays, enhances security through authentication mechanisms, and enables real-time tracking of loan applications. Testing results confirm that the system functions as expected under normal and boundary conditions.

The project was completed within the planned academic timeline and demonstrates the practical application of theoretical concepts learned in Software Engineering and Web Development courses.

Future enhancements may include mobile application support, AI-based credit scoring, and integration with additional banking services.
